M.SUBHI ABO RDAN

+1(857)-693-9726 msubhi_a@mit.edu LinkedIn GitHub Portfolio Google Scholar

Education

Massachusetts Institute of Technology (MIT) Cambridge, MA

Master of Engineering in Electrical Engineering and Computer Science (MEng) *June 2027*

Bachelor of Science in Computer Science and Engineering — Physics *June 2026*

- **MEng Advisors:**
  Prof. Philip Harris and Prof. Hengyun Zhou
- **MEng Research:**
  Real-Time Learning and FPGA Acceleration for Quantum Control and Quantum Error Correction
- **EECS/Systems Coursework:**
  Digital Systems Laboratory II (FPGA), Accelerated Computing (GPUs), Software Performance Engineering, Computer Architecture, Distributed Systems, Operating Systems, Complexity and Computability Theory.
- **Physics/Theory Coursework:**
  Quantum Information Science II, Quantum Error Correction, Quantum Computation, Quantum Engineering Platforms, Quantum Mechanics II, General Relativity, Statistical Mechanics, Deep Learning, Reinforcement Learning, Statistical Data Analysis.

Research and Work Experience

Optiver June 2026 – August 2026

FPGA Engineering Intern *Chicago, IL*

- FPGA Acceleration for High Frequency Trading

Apple May 2025 – August 2025

Silicon Prototyping Engineering Intern — FPGA Engineer *Cupertino, CA*

- Built a non-invasive distributed clock monitoring system for multi-board FPGA setups, enabling run-time frequency measurement and verification of external clocks via a register-mapped UMRBus3 interface.
- Engineered a cycle-accurate FPGA validation engine using RTL waveform replays to emulate the rest of the system, enabling high-throughput DUT testing.
- Designed a software/hardware execution pipeline to parse and compress VCD transition traces (220× reduction) and drive FPGA-based DUT validation.

MIT Space Systems Laboratory — STAR Lab June 2024 – August 2025

Research Assistant — FPGA Engineer *Cambridge, MA*

- Designed, implemented, and performance-tested an optical modeling system for high-order wavefront sensing and control algorithms on the AMD Zynq UltraScale+ MPSoC ZCU104 using High-Level Synthesis (HLS).
- Contributed to a **first-author publication at SPIE Photonics West 2025**: "Accelerating High-Order Wavefront Sensing and Control Algorithms with FPGAs".
- Designed a 2D FFT accelerator, achieving a 360× improvement in execution time over the Arm Cortex A53 CPU.
- Developed an Angular Spectrum Method accelerator interfacing with the CPU via Xilinx Runtime (XRT) C++ API, optimizing dataflow, reducing data movement, and increasing parallelism, achieving a 500× speedup over CPU.

MIT Multifunctional Metamaterials — META Lab Feb 2024 – Jun 2024

Research Assistant *Cambridge, MA*

- Developed a physical model to investigate the snap-in dynamics of microcantilevers in Atomic Force Microscopes.
- Constructed a computational simulation utilizing finite difference method. This simulation predicts the optimal snap-in time and distance critical for achieving high-resolution nano-scale measurements in AFMs.

MIT Media Lab – Signal Kinetics Jun 2023 – Sep 2023

Research Assistant *Cambridge, MA*

- Designed a simulation using Python based on the Jaffe and McGlamery underwater imaging model to generate underwater images of the Macbeth ColorChecker that accurately replicates real-world underwater conditions.
- Developed an algorithm capable of reconstructing color images by integrating information from multiple monochromatic underwater images taken at different distances. On going evaluation with real data.

Projects

Intermediate-Code Automorphisms in Pieceable Fault-Tolerant Quantum Computation May 2026

- Developed a framework for studying dynamically evolving stabilizer-code symmetries during pieceable fault-tolerant quantum gate constructions, focusing on logical Clifford and multi-qubit gate synthesis.
- Implemented automorphism-search and logical-action verification tools using Python, Stim, and GF(4)/symplectic representations to analyze intermediate quantum error-correcting codes and encoded logical operations.

Near-Memory FPGA 100G-Ethernet UDP Engine December 2025

- Designed and implemented a low-latency, line-rate 100 Gbps UDP offload engine on an AMD UltraScale+ RFSoC operating at 322.22 MHz, including connection bookkeeping via a fully pipelined, set-associative hash table.
- Built a SmartNIC-style near-memory architecture leveraging AXI HP and cache-coherent ACP ports via DMA, enabling ultra-low-latency packet delivery directly into CPU memory and caches.

Fault-Tolerant Sharded Key-Value Store April 2025

- Implemented the Raft distributed-systems consensus algorithm. Integrated snapshotting to compress Raft logs, reducing recovery latency and minimizing storage footprint.
- Built a distributed, fault-tolerant key-value store with dynamic sharding using Raft, supporting reconfiguration and load balancing via RPC-based coordination among cluster nodes.

FPGA QR-Code Reader December 2023

- Developed a real-time QR code scanner on an AMD Urbana FPGA, integrating an OV7670 camera module and HDMI output; awarded the **MIT EECS George C. Newton Laboratory Prize** (2024).
- Implemented image processing pipelines to grayscale, denoise, detect alignment patterns, and decode QR codes as small as 240×240 pixels in under 0.02 seconds.

Publications

- **M. Subhi Abo Rdan**, N. Belsten, S. Rao, A. W. Taka, B. Eickert, K. Milani, E. S. Douglas, L. Pogorelyuk, K. Cahoy, "Accelerating high-order wavefront sensing and control algorithms with FPGAs," Proc. SPIE 13627, 2025. DOI: 10.1117/12.3064675
- B. Eickert, N. Belsten, K. Milani, A. Taka, **M. Subhi Abo Rdan**, L. Pogorelyuk, S. Rao, E. S. Douglas, K. Cahoy, "Demonstrating real-time wavefront sensing and control with space processors," Proc. SPIE 13627, 2025. DOI: 10.1117/12.3065650
- P. Nair, C. Panagiotou, M. Masterson, K. De, E. Kara, E. Winkler, **M. Subhi Abo Rdan**, "A Suppressed Volumetric Rate of High-Luminosity Mid-Infrared Selected Tidal Disruption Events," accepted to *The Astrophysical Journal*, 2026. arXiv:2606.31926.

Honors & Awards

- MIT EECS George C. Newton Laboratory Prize, 2024
- International Physics Olympiad (IPhO), Bronze Medalist, 2020 & 2021
- European Physics Olympiad (EuPhO), Bronze Medalist, 2020 & 2021
- International Applied Physics Olympiad (IAPhO), Silver Medalist, 2021
- Physics Cup, Honorable Mention, 2021

Skills and Interests

Languages: C/C++, SystemVerilog, VHDL, Python, Go, CUDA, TypeScript, RISC-V Assembly, TCL.

Hardware & Systems: FPGA Design, High-Level Synthesis (HLS), Computer Architecture, Distributed Systems, Networking, SmartNICs, GPU Computing, Quantum Error Correction.

Tools/Frameworks: Vivado, Vitis, XRT, ProtoCompiler, Cocotb, Pynq, Stim, Git, Linux/UNIX.

Other Experience: FPGA Teaching Assistant, Quantum Physics Grader, Citadel Securities Quantitative Invitational.

Activities: Weight Lifting, Running, Ice Hockey, Tennis, Swimming, Soccer.